

6.6

12. $M = I$

$JM = J \neq MK = K$

Alternatively, the δ matrix gives:

$M = \delta_{n \times n}$

$\Rightarrow JM = O = MK = O$

But $M = \delta_{n \times n}$ is not ∇ invertible

$\therefore M^{-1}JM = K$ is impossible

Ehh...

More info?

20. a. $A^k = MJ^kM^{-1}$

$\Rightarrow B^k = MJ^kM^{-1}$

$\therefore$ Changes apply only to the Jordan matrix J . Therefore the set of matrices remain "constant".

e. $M = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

22. A has all $\lambda = 0$

Since we think of Jordan blocks, we're really saying that as we take higher and higher powers, the Jordan blocks will "cancel out" to produce 0's even on upper-diagonal entries.

6.7

11. Since A is already orthogonalized and length $\delta_1, \dots, \delta_n$, then $V = I$ and as such $\delta = \begin{bmatrix} \delta_1 & & \\ & \ddots & \\ 0 & & \delta_n \end{bmatrix}$ with corresponding rows from A on V .

14. Let $J_c = \begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}$

1. $M_c^{-1}J_cM_c = J_c^T$
 $\Rightarrow J_cM_c = J_c^TM_c$

$\Rightarrow J_c^T = \begin{bmatrix} 4 & 0 \\ 1 & 4 \end{bmatrix}$

$\Rightarrow M_c = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \Rightarrow M_c^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

2. $J^T = \begin{bmatrix} M_1^{-1} & & \\ & M_2^{-1} & \\ & & \ddots \\ & & & M_n^{-1} \end{bmatrix} J \begin{bmatrix} M_1 & & \\ & M_2 & \\ & & \ddots \\ & & & M_n \end{bmatrix}$
 $= \begin{bmatrix} M_1^{-1}JM_1 & & \\ & M_2^{-1}JM_2 & \\ & & \ddots \\ & & & M_n^{-1}JM_n \end{bmatrix} = J^T$

3. $A^T = (MJM^{-1})^T = M^{-1T}J^TM^T = (M^T)^{-1}J^TM^T$

$\therefore A^T$ is similar to J^T ; J^T is similar to J and J is similar to A .

b. Perhaps J included some $\lambda_i < 0$ so that all $\lambda_i \geq 0$ now the squared-mass removes these negative evlues.

c. These contain non-repeated evlues $\Rightarrow$ some true, det, and evectors $\Rightarrow$ similar.

d. Though some evlues and such, these do not contain similar evectors as cI for any c is only similar to itself.

23. $A_1 = Q_1R_1-cs^2I$
 $A_2 = Q_2R_2-cs^2I$

$\Rightarrow Q_1A_2Q_1^{-1}$
 $= Q_1(R_2-cs^2I)Q_1^{-1}$
 $= Q_1R_2-cs^2Q_1Q_1^{-1}$
 $= Q_1R_2-cs^2I = A_1$

Ehh...

(Good enough)

24. If A is similar to A^{-1} , $A = MJM^{-1} = A^{-1}$ must hold. Remember if all $\lambda_i = 1$ or -1 so long invertibility holds.

Technically I think this is fine enough.

17. a. First assume

$x^3 - 6x^2 + 10x - 4$
 $\Rightarrow (x-2)(x^2 - 4x + 2)$

Therefore the roots

are $\{ \frac{1}{\sqrt{2}}, [0, 1], [-\frac{1}{\sqrt{2}}] \}$

b. $AV = \frac{1}{2} \begin{bmatrix} 1 & \sqrt{2} & \sqrt{2} \\ \sqrt{2}-1 & -\sqrt{2} & -\sqrt{2}-1 \\ -\sqrt{2} & \sqrt{2} & 1+\sqrt{2} \\ -1 & \sqrt{2} & -1 \end{bmatrix}$

c. $A^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$

$\therefore N(A^T) = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

8.5

$$4. \int_{-1}^1 x^3 - cx \, dx = 0, \int_{-1}^1 x^4 - cx^2 \, dx = 0 \quad \int_{-1}^1 (x^3 - cx)(x^2 - \frac{1}{3}) \, dx = 0$$

Nice comeback

* Notice that the only commonality

between these three is that

$$c = x^2$$

However, we could insert it in a polynomial:

$$\frac{1}{5}x^4 - \frac{c}{3}x^2 \Rightarrow 0 \Rightarrow \frac{2}{5} - \frac{2}{3}c$$

$$\therefore c = \frac{3}{5}$$

$$5. \int_0^{2\pi} f(x) \cos x \, dx = 0$$

$$\Rightarrow \int_0^{\pi} \cos x \, dx - \int_{\pi}^{2\pi} \cos x \, dx = 0$$

$$\int_0^{2\pi} f(x) \sin x \, dx = 4$$

$$\Rightarrow \int_0^{\pi} \sin x \, dx - \int_{\pi}^{2\pi} \sin x \, dx = 4$$

$$\int_0^{2\pi} f(x) \sin 2x \, dx$$

$$= \int_0^{\pi} \sin 2x \, dx - \int_{\pi}^{2\pi} \sin 2x \, dx = 0$$

$$12. \frac{d}{dx}(1) = 0, \quad \frac{d}{dx}(\cos x) = -\sin x, \quad \frac{d}{dx}(\sin x) = \cos x, \quad \frac{d}{dx}(\cos 2x) = -2\sin 2x, \quad \frac{d}{dx}(\sin 2x) = 2\cos 2x$$

$$\Rightarrow \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 2 & 0 \end{bmatrix} \quad h/2$$

$$13. a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \, dx = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} 1 \, dx = 1/\pi$$

$$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos kx \, dx \Rightarrow \left. \frac{1}{\pi k} \sin kx \right|_{-\pi/2}^{\pi/2} = \left(\frac{2}{\pi k} \sin\left(\frac{k\pi}{2}\right) \right)$$

$$b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin kx \, dx \Rightarrow \left. \frac{1}{\pi k} \cos kx \right|_{-\pi/2}^{\pi/2} = 0$$

Woo-hoo! Last part!